Articulatory Data Transformation & Rigid Body Registration

Gordon L. Mills
Department of Psychology & Language Sciences, University College London

November 3, 2026

Abstract

This note describes the mathematics employed in ema-trans.py, a tool for performing bite plane transformation and rigid body registration on electromagnetic articulography (EMA) data.

1 Bite Plane Transformation

EMA sensor data are referenced to a right-handed Cartesian coordinate system (x, y, z). The origin is located at the centre of the field generator unit (FGU). The **x** and **y** axes lay vertically and horizontally in the FGU face plane, with the **z** axis normal to the plane. A bite plane is defined with a right-handed Cartesian coordinate system (i, j, k). The origin is located at the central maxillary incisors. The **i** and **k** axes lay sagittally and transversely in the occlusal plane, with the **j** axis normal to the plane. We seek a transformation from the EMA coordinate system to the bite plane coordinate system.

Three reference sensors, located at the nasion (A), left mastoid (B), and right mastoid (C) are employed during data collection. Three additional sensors, located at the top (F), left (G), and right (H) of a bite plate are employed during a bite plate recording. The mean position of the point A is computed as,

$$\overline{\mathbf{a}} = \frac{\sum_{n=0}^{N-1} w_n \mathbf{a}_n}{\sum_{n=0}^{N-1} w_n}, \tag{1}$$

where w_n is unity if all six of the sensor data points are valid for sample n, and zero otherwise. The mean positions of the remaining reference and bite plate sensors are computed, yielding the vectors $\overline{\mathbf{a}}, \overline{\mathbf{b}}, \overline{\mathbf{c}}, \overline{\mathbf{f}}, \overline{\mathbf{g}}, \overline{\mathbf{h}}$.

The translated mean mid-point of the bite plate left and right sensor vectors is computed,

$$\overline{\mathbf{m}} = \frac{\overline{\mathbf{g}} + \overline{\mathbf{h}}}{2} - \overline{\mathbf{f}}, \tag{2}$$

yielding a vector lying sagitally in the occlusal plane. The vector $\overline{\mathbf{a}} - \overline{\mathbf{f}}$ lies in the sagittal plane, hence the cross-product of these vectors yields an orthogonal vector from the occlusal origin in the **k** direction. Vectors from the occlusal origin in the **j** and **i** directions are computed similarly,

$$\mathbf{k} = (\overline{\mathbf{a}} - \overline{\mathbf{f}}) \times \overline{\mathbf{m}}, \qquad \mathbf{j} = \overline{\mathbf{m}} \times \mathbf{k}, \qquad \mathbf{i} = \mathbf{j} \times \mathbf{k}. \tag{3}$$

The normalisation of each of these vectors form a row of the required rotation matrix,

$$\mathbf{R_b} = \begin{pmatrix} \hat{i}_x & \hat{i}_y & \hat{i}_z \\ \hat{j}_x & \hat{j}_y & \hat{j}_z \\ \hat{k}_x & \hat{k}_y & \hat{k}_z \end{pmatrix}. \tag{4}$$

Each of the mean reference vectors $\overline{\mathbf{a}}, \overline{\mathbf{b}}, \overline{\mathbf{c}}$ form a column of a matrix, which is translated to the occlusal origin before being multiplied by the rotation matrix, yielding the transformed mean reference matrix,

$$\mathbf{Q} = \mathbf{R_b} \left[\begin{pmatrix} \overline{a}_x & \overline{b}_x & \overline{c}_x \\ \overline{a}_y & \overline{b}_y & \overline{c}_y \\ \overline{a}_z & \overline{b}_z & \overline{c}_z \end{pmatrix} - \overline{\mathbf{f}} \right] = \begin{pmatrix} \overline{a}_i & \overline{b}_i & \overline{c}_i \\ \overline{a}_j & \overline{b}_j & \overline{c}_j \\ \overline{a}_k & \overline{b}_k & \overline{c}_k \end{pmatrix}. \tag{5}$$

Subsequent rigid body registration to this matrix, from recorded reference sensor data, additionally transforms the point data.

2 Rigid Body Registration

If bite plane transformation was not utilised, the mean positions of the reference sensors, located at the nasion (A), left mastoid (B), and right mastoid (C), are computed. This is performed using Equation (1), where w_n is unity if all three of the sensor data points are valid for sample n, and zero otherwise. Each of the resulting mean reference vectors $\overline{\mathbf{a}}, \overline{\mathbf{b}}, \overline{\mathbf{c}}$ form a column of the mean reference matrix,

$$\mathbf{Q} = \begin{pmatrix} \overline{a}_x & \overline{b}_x & \overline{c}_x \\ \overline{a}_y & \overline{b}_y & \overline{c}_y \\ \overline{a}_z & \overline{b}_z & \overline{c}_z \end{pmatrix}. \tag{6}$$

In all cases, the mean of the columns of $\mathbf{Q}$ is then computed, yielding the mean reference centroid $\overline{\mathbf{q}}$, and the following operations performed for samples $n = 0, 1, \ldots, N-1$: The sample reference matrix is formed from the sample reference vectors,

$$\mathbf{P} = \begin{pmatrix} a_x & b_x & c_x \\ a_y & b_y & c_y \\ a_z & b_z & c_z \end{pmatrix}, \tag{7}$$

and the mean of the rows of $\mathbf{P}$ computed, yielding the sample reference centroid $\overline{\mathbf{p}}$. A covariance matrix is calculated from the centred sample and mean reference matrices,

$$\mathbf{H} = (\mathbf{P} - \overline{\mathbf{p}})(\mathbf{Q} - \overline{\mathbf{q}})^{\mathsf{T}}, \tag{8}$$

and the singular value decomposition satisfying $\mathbf{H} = \mathbf{U}\mathbf{S}\mathbf{V}^{\mathsf{T}}$ is computed. The optimal rotation matrix is calculated from the found unitary matrices,

$$\mathbf{R_r} = \mathbf{V} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \det(\mathbf{V}\mathbf{U}^{\mathsf{T}}) \end{pmatrix} \mathbf{U}^{\mathsf{T}}. \tag{9}$$

The optimal translation vector,

$$\mathbf{t} = \overline{\mathbf{q}} - \mathbf{R_r}\overline{\mathbf{p}}, \tag{10}$$

is calculated. Finally, the affine transformation,

$$\mathbf{p'}_m = \mathbf{R_r}\mathbf{p}_m + \mathbf{t}, \tag{11}$$

is applied to point data $\mathbf{p}_m$ from sensors $m = 0, 1, \ldots, M-1$.

References

[1] Sorkine-Hornung & Rabinovich (2017) *Least-Squares Rigid Motion Using SVD.* Retrieved June 25, 2025, from `https://igl.ethz.ch/projects/ARAP/svd_rot.pdf`